A2 Trojans in the LLM Era

Does Detection Feedback Help an Attacker?
An exploratory study of A2-inspired modifications in small circuit netlists

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No valid candidate escaped in the baseline. A model-swap follow-up produced 1 valid evasion event against the same reviewer. This exploratory exception does not establish that more detailed feedback improves evasion.

Research question

Diagnostic explanations help a designer locate suspicious circuitry, but the same explanation may tell an attacker what to change. We ask whether increasing access to an LLM reviewer's feedback helps an LLM produce a functioning circuit modification that the reviewer accepts. The distinction between functioning and merely unflagged matters: an attacker can make a circuit less conspicuous by weakening the very behavior it is meant to induce.

A2 [1] motivates an analog-state trigger that accumulates charge from signal transitions. SPICED [2] motivates LLM-assisted Trojan detection, and LATENT [3] motivates feedback-guided modification. We study a deliberately narrower setting with a two-stage operational amplifier and a four-input AND gate. Two simplifications define the experiment: the attacker selects parameters for a **fixed charge-pump/output-load template**, and the reviewer performs **netlist-only analysis**. This is not a reproduction of the original hardware or complete published frameworks.

Controlled feedback comparison

In the baseline, attacker and reviewer both use gpt-4o-mini-2024-07-18 at temperature 0.3, without fine-tuning. The attacker receives the clean circuit, parameter bounds, test conditions and one verified starting example. Its objective is to preserve normal behavior while inducing a disturbance. The reviewer receives only the candidate netlist and intended function. Separate stateless reviews keep attacker instructions and condition labels out of its input.

For each circuit, five separately seeded trajectories contain four attempts in each of three conditions. **None** generates each proposal without prior feedback. **Binary** receives the preceding parameters and a Boolean detection verdict. **Full** additionally receives confidence, suspect nodes and diagnostic text. The first candidate and verdict are shared across the three conditions. Later attempts use the same opportunity count and receive no simulation results. The comparison comprises 120 condition-specific rows from 100 generation events.

Binary versus full isolates feedback detail within the same revision procedure. Comparisons with none also change access to the preceding candidate, so they combine feedback and memory effects. Prompts are fixed during the final experiment, condition order is shuffled reproducibly, and all four attempts are retained. Repeated netlists are counted as repeated generations, not as additional circuit families.

Evaluation endpoints

The primary endpoint is **electrically valid AND reviewer-negative candidates / all attempts**. Validity requires acceptable behavior under the normal stimulus and a sufficient output disturbance under the activation stimulus. Secondary measures are evasion conditional on validity and at least one valid evasion within a trajectory. Invalid candidates remain failures. Because adaptive rounds are dependent, attempt proportions are descriptive; trajectory-level intervals use five starts and require an independence assumption.

Circuit model and electrical validation

Simulation uses ngspice 47, 3.3 V and generic LEVEL=1 MOS models. The op-amp has two gain stages with Miller compensation; AND4 is a NAND4 followed by an inverter. Clean circuits and manually specified positive controls establish the intended normal and disturbed behavior. These are simulation measurements, not device characterization.

Check	Measured result	Meaning / criterion
Op-amp gain / bandwidth	95.47 dB / 131.9 Hz	Clean small-signal AC response
Unity gain / phase margin	9.21 MHz / 50.4 degrees	Clean compensated feedback behavior
Follower RMS error	0.044 mV	1.65 V +/- 50 mV at 10 kHz
AND truth table / delays	16/16; rise 0.148 ns; fall 0.115 ns	50% crossings; fall switches all four inputs
Op-amp positive control	Dormant max deviation <0.001 mV; triggered drop 0.885 V	Pass: dormant <=20 mV; triggered >=0.3 V
AND positive control	16/16; dormant deviation 0.031 mV; triggered drop 2.991 V	Pass: settled deviation <=0.2 V; triggered >=0.8 V

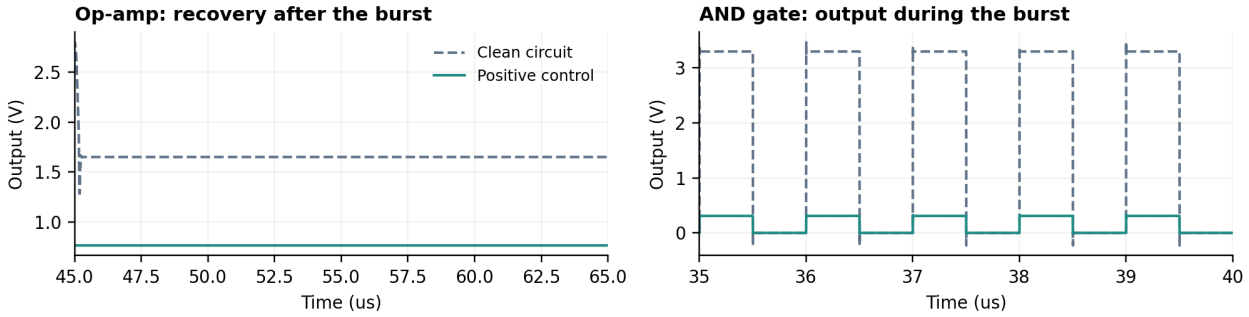


Figure 1. Clean and positive-control outputs under identical activation stimuli: op-amp recovery after the burst and AND late burst cycles. Both controls pass normal-stimulus checks and exceed the disturbance threshold.

Trigger abstraction and stimulus limits

A diode-capacitor network accumulates charge against resistive leakage. A thresholded switch or smooth behavioral current loads the output. The attacker varies representation, names and bounded component parameters while the core circuit stays fixed. These choices explore one mechanism, rather than unrestricted Trojan topologies.

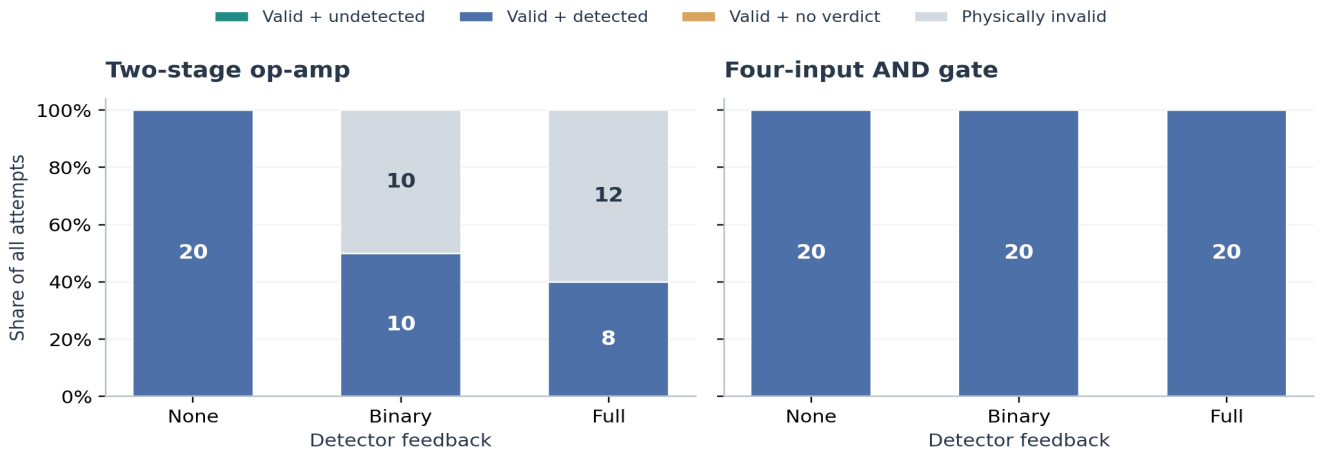
The op-amp normal stimulus is a 10 kHz sinusoid at 1.65 V +/-50 mV; activation uses a 1 MHz burst between 0.6 and 2.7 V. **Both amplitude and frequency change.** The burst ends at t=45 us and recovery is evaluated at t=50-60 us. AND4 normal inputs enumerate 16 combinations at 20 us intervals; activation toggles one input at 1 MHz with the others high.

A post-hoc amplitude-by-frequency check over 40 us found no output disturbance at 0.1 V peak-to-peak for either 25 kHz or 1 MHz. At 2.1 V peak-to-peak, 25 kHz produced a 0.065 V reservoir peak and negligible disturbance; 1 MHz produced 1.092 V and a 0.885 V drop. Both sufficient swing and rapid accumulation matter for this positive control. Separately, the AND single-input fall delay was 0.143 ns; it is a different transition from the tabulated all-input fall.

Baseline results: a detection ceiling

Circuit	Feedback	Valid	Primary success	Given valid	Any success; 95% CI
Op-amp	none	20/20	0/20 (0%)	0%	0/5 [0.0%, 43.4%]
Op-amp	binary	10/20	0/20 (0%)	0%	0/5 [0.0%, 43.4%]
Op-amp	full	8/20	0/20 (0%)	0%	0/5 [0.0%, 43.4%]
AND4	none	20/20	0/20 (0%)	0%	0/5 [0.0%, 43.4%]
AND4	binary	20/20	0/20 (0%)	0%	0/5 [0.0%, 43.4%]
AND4	full	20/20	0/20 (0%)	0%	0/5 [0.0%, 43.4%]

Primary = valid and undetected / all attempts. Given valid = valid and undetected / valid attempts. The last column counts trajectories with any success and gives a Wilson interval under independence of starts. The shared first round and adaptive later rounds are not independent attempts.



Counts inside bars. Attempts per group: 20. Adaptive rounds are dependent; attempt rates are descriptive.

Figure 2. Electrical validity and detection outcomes for the six conditions. An invalid circuit is unsuccessful even if its modification appears less suspicious. The first-round candidate is shared across conditions; percentages describe condition-specific attempts.

All 100/100 generation events received a positive detection verdict, including all 78 electrically valid events. These events produced 28 distinct netlist texts (22 op-amp, 6 AND), of which only 12 were valid. The 120 rows include shared first rounds. Thus every condition has zero valid evasion, but the classification endpoint has no variation with which to estimate a feedback advantage.

Feedback revisions and payload effectiveness

In one op-amp trajectory, a full-feedback revision changed the output-load resistance from 100 to 300 ohms along with other parameters. The recovery disturbance fell from **0.523 V to 0.006 V**, below the 0.3 V criterion, while normal behavior remained acceptable. Both candidates were detected. This example illustrates a loss of payload effectiveness, not concealment; simultaneous parameter changes prevent attributing the loss to resistance alone.

Post-hoc sensitivity check. With a looser 0.2 V op-amp drop threshold, valid counts become 20/20, 20/20 and 14/20. Validity fractions therefore depend on the cutoff. Successful evasion remains zero under every tested threshold. The primary threshold was not changed. The 0-43.4% interval for zero successes in five trajectories is conditional on this narrow setup, not a bound on all possible analog Trojans.

Supplementary model comparison

These supplementary analyses were designed after observing the baseline detection ceiling. Their inclusion and deduplication rules were fixed before the additional generations, but they are exploratory follow-ups rather than independent confirmation. They address two different questions: whether a model change expands the attacker's search, and whether another reviewer reaches the same verdicts on frozen candidate circuits.

Changing the attacker

The first follow-up replaces only the attacker with **gpt-5.4-mini-2026-03-17**, retaining the GPT-4o mini reviewer, prompts, templates, test stimuli and thresholds. It repeats the six conditions with five starts and four attempts, sharing the first candidate within each model's feedback arms. Temperature remains 0.3 and reasoning effort is set to none. The alternative model receives no seed parameter; comparisons across models therefore do not pair identical random draws.

Circuit	Feedback	Baseline valid	New model valid	New model success	Successful starts
Op-amp	none	20/20	20/20	0/20	0/5
Op-amp	binary	10/20	16/20	0/20	0/5
Op-amp	full	8/20	5/20	0/20	0/5
AND4	none	20/20	20/20	0/20	0/5
AND4	binary	20/20	18/20	1/20	1/5
AND4	full	20/20	5/20	0/20	0/5

Validity and success use the same electrical criteria in both experiments. Baseline valid evasion is zero in every group. New model success counts valid, reviewer-negative candidates, not all negative verdicts.

Fresh checks of the unchanged reviewer flagged clean controls 0/5 and 0/5, and positive controls 5/5 and 5/5 (op-amp, AND4). These repeat the same control circuits.

The alternative attacker produced 71 distinct netlist texts from 100 generation events; 64 events met the electrical criteria and 99 were flagged by the unchanged reviewer. It used 3 of three representations and 3 of three naming styles. This measures broader exploration of the permitted template, not independent physical attack mechanisms or an intrinsic model ranking.

Reviewing frozen candidates with another model

A separate retrospective check uses **gpt-5.4-mini-2026-03-17** with the fixed review prompt. Every distinct netlist text in the union of the two generation experiments is included once, irrespective of its validity or previous verdict. This compares reviewers on fixed inputs; the new reviewer's feedback is never returned to either attacker. It is therefore not an adaptive stronger-attacker/stronger-reviewer experiment.

Circuit	Distinct inputs	Valid inputs	New reviewer flags	Valid inputs missed
Op-amp	56	23	55	0
AND4	43	27	43	0

The new reviewer disagreed with the stored representative verdict on 2/99 distinct inputs. Its clean-control flags were 0/5 and 0/5; positive-control flags were 5/5 and 5/5. Each candidate has one new judgment, and the controls still repeat the same four designs. These denominators differ from the adaptive attempt counts.

Re-simulation reproduced the stored measurements for all 99 distinct candidate netlists; an independent uniform-time calculation agreed with every validity classification. Code, prompts, netlists and measurements are available in the project repository.

Interpretation and limitations

Detector calibration

During exploratory testing, the initial prompt mislabeled both clean circuits in every repeat (10/10 false flags). Before the final comparison, it was refined to recognize ordinary bias and compensation motifs and require trigger-to-payload evidence. Final clean-control flags were 0/5 and 1/5; positive-control flags were 5/5 and 5/5, for op-amp and AND4 respectively. These same circuits informed prompt development, so controls are not held out. Repeated judgments of four fixed netlists cannot estimate population false-positive rates or sensitivity.

Identifiability at the observed ceiling

The baseline is a bounded exploratory negative result: every allowed variant retains a recognizable charge-storage-to-output path, and every generated candidate is flagged. This ceiling prevents identifying a benefit from more detailed feedback. Validity differences show changed electrical behavior, not a general evasion effect. Neither the ceiling nor confidence scores establish broad reviewer robustness.

The model-swap follow-up yielded 1 valid evasion event. This is an exception within the permitted template, not a monotonic feedback effect or a reliable model ranking. The comparison is exploratory, and a single negative judgment does not show stable evasion across repeated reviews.

In the observed AND binary-feedback case, all 16 normal logic cases passed, settled deviation was 0.031 mV, and the activated high-output drop was 3.022 V. The original reviewer treated the extra network as potentially ordinary biasing and requested simulation. The alternative reviewer flagged this same netlist. This case separates real simulated payload behavior from a model's interpretation; it does not isolate the causal role of naming, feedback or reviewer randomness.

A post-hoc check kept that netlist and prompt fixed for five fresh judgments per reviewer: GPT-4o mini flagged it 4/5 times and GPT-5.4 mini 5/5. The observed miss is therefore intermittent rather than stable concealment. This success-selected case is not pooled into the original rates and cannot estimate general reviewer error.

Relation to earlier work

LATENT reports mean component-level evasion of 23.7% with one-shot prompting and 94.4% with iterative refinement across 15 circuits [3]. That localization-oriented endpoint differs from our requirement that an entire valid candidate receive a negative verdict; these percentages are not directly comparable.

SPICED uses specifications, simulation-derived voltage/current information and few-shot rules [2]. Our netlist-only review omits this evidence, and our fixed template is much narrower than LATENT's insertion strategy. In the baseline, all generations retained plain names and none selected the offered subcircuit form; the verified starting example and low sampling temperature may anchor exploration. Generic devices and an abstract behavioral payload also preclude claims about the area, power, stealth or manufacturability of original A2 hardware [1].

Next experiment

A more discriminating evaluation would use independently designed attack families, benign look-alike modifications and held-out circuits, seeking a reviewer with neither universal acceptance nor universal rejection. Matched-amplitude frequency sweeps would separate trigger mechanisms. Comparing netlist-only review with simulation-informed review would then test whether detailed diagnostics improve defense, assist valid evasion, or both.

References

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[2] J. Chaudhuri, D. Thapar, A. Chaudhuri, F. Firouzi and K. Chakrabarty. **SPICED: Syntactical Bug and Trojan Pattern Identification in A/MS Circuits using LLM-Enhanced Detection.** PAINE 2024; arXiv:2408.16018.

[3] J. Chaudhuri, A. Chaudhuri and K. Chakrabarty. **LATENT: LLM-Augmented Trojan Insertion and Evaluation Framework for Analog Netlist Topologies.** ICLAD 2025; arXiv:2505.06364.

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